

POTHOLELOG · DESIGN PRECIS

PotholeLog design precis

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	PHL-PRC-001	Version	0.2
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/potholelog		
Source	docs/02-concept.md @ 583c574		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, safety, media)
0.1	2026-09-25	Amish Chadha	Initial scaffold

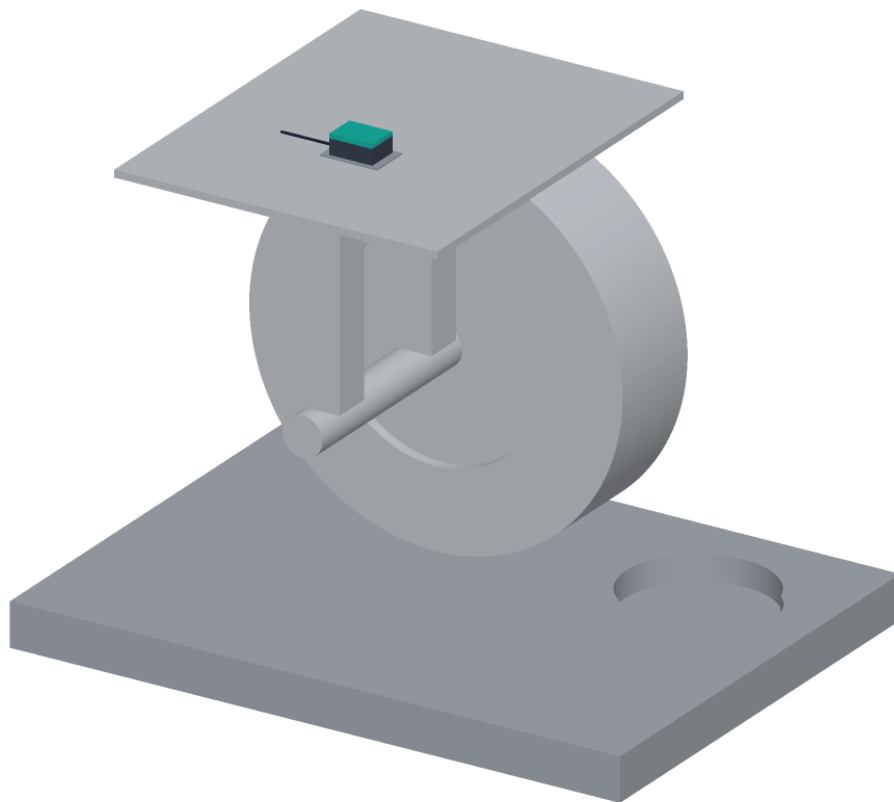
PotholeLog design precis

■ Summary

PotholeLog is a sealed box, about 120 x 90 x 59 mm, bolted to the floor of a bus or refuse truck above the rear axle. It records how the vehicle shakes and where it is, turns each 100 m of road into a roughness value and flags sharp impacts as possible potholes, then uploads a small daily file over depot Wi-Fi. Averaged over many passes and calibrated per vehicle, the data gives a city a weekly map of road condition on every street its fleet drives. First-order numbers suggest off-the-shelf modules meet the sampling, storage and power needs for about \$69 in parts, just inside the \$70 budget. Detection accuracy and calibration to IRI are unverified until field data exist.

PotholeLog

CONCEPT, NOT FOR FABRICATION



Grey: Road with pothole, Wheel, axle, suspension and floor section for scale

Figure 1. Concept massing model. The logger is the small dark box on the floor section; the grey scene is for scale.

■ How it works

1. **Sense.** A 6-axis IMU rigidly fixed to the enclosure floor samples acceleration at 400 Hz. A GNSS module under the lid logs position and speed at 10 Hz.
2. **Log raw data.** Raw samples go to a microSD card as a rolling buffer, so any detection can be re-run later with a better algorithm.

3. **Summarize on board.** For every 100 m traveled, the controller computes a roughness value from the vertical acceleration (for example, band-limited RMS normalized for speed), and records the segment's start and end, speed and the number of samples. Sharp vertical impacts above a threshold are stored as defect events with position, speed and peak size.
4. **Upload.** When the vehicle is back at the depot and the ignition is switched off or on, the controller joins the depot Wi-Fi and uploads the day's summaries to the fleet's server.
5. **Calibrate and aggregate.** On the server, each vehicle's raw roughness is converted to an IRI estimate using a calibration equation found on reference sections, in the way World Bank Technical Paper 46 describes for response-type systems (Sayers et al., 1986). Events from several passes and vehicles are clustered; a cluster seen on repeated passes becomes a reported defect, as in Pothole Patrol (Eriksson et al., 2008).
6. **Publish.** The city publishes a map of segment roughness and defect clusters as CSV and GeoJSON, for its own GIS and for CityTwin. Vehicle tracks and timestamps stay on the operator's server.

PotholeLog: data flow (estimates; road data only, no images or audio)
CONCEPT, NOT FOR FABRICATION

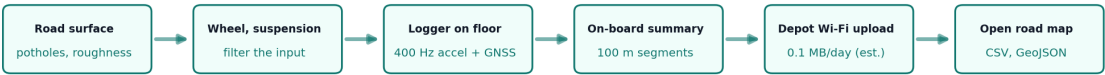


Figure 2. Data flow. Values are estimates.

■ Main components

Table 1. Main components. Numbers match the exploded view (Figure 3) and bom/bom.csv .

#	Component	Proposed choice	Notes
1	Mounting plate	160 x 110 x 4 mm aluminum, 4 x M6	Bolts to existing floor or seat-rail fixings; rigid so the IMU sees the vehicle, not the bracket
2	Enclosure base	Stock IP65 ABS or polycarbonate box, 120 x 90 mm	IMU screwed directly to its floor
3	Enclosure lid	Supplied with the box, with gasket	GNSS antenna under the lid; plastic lid keeps sky view through vehicle windows
4	DC-DC converter and protection	9 to 36 V in, 5 V 1 A out; TVS diode, reverse-polarity diode, input fuse	Covers 12 V and 24 V vehicles
5	Hold-up supercapacitor	1 F, 5.5 V	Keeps the controller alive long enough to close files when power drops
6	Controller	ESP32-S3 board with Wi-Fi and microSD slot	Proposed, awaiting Amish (alternatives below)
7	IMU	6-axis, LSM6DSO class, ± 16 g, 400 Hz or more	Gyroscope helps separate body roll and pitch from vertical motion
8	GNSS module	u-blox M10 class with patch antenna, 10 Hz	Multi-constellation
9	Cable gland and fused lead	M16 gland, 2 m lead, inline 2 A fuse	Wired to an ignition-switched fused circuit

PotholeLog: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Mounting plate, aluminum
- 2 Enclosure base, IP65
- 3 Enclosure lid with gasket
- 4 DC-DC converter and protection
- 5 Hold-up supercapacitor
- 6 Controller, Wi-Fi, microSD
- 7 6-axis IMU
- 8 GNSS module and patch antenna
- 9 Cable gland and fused lead

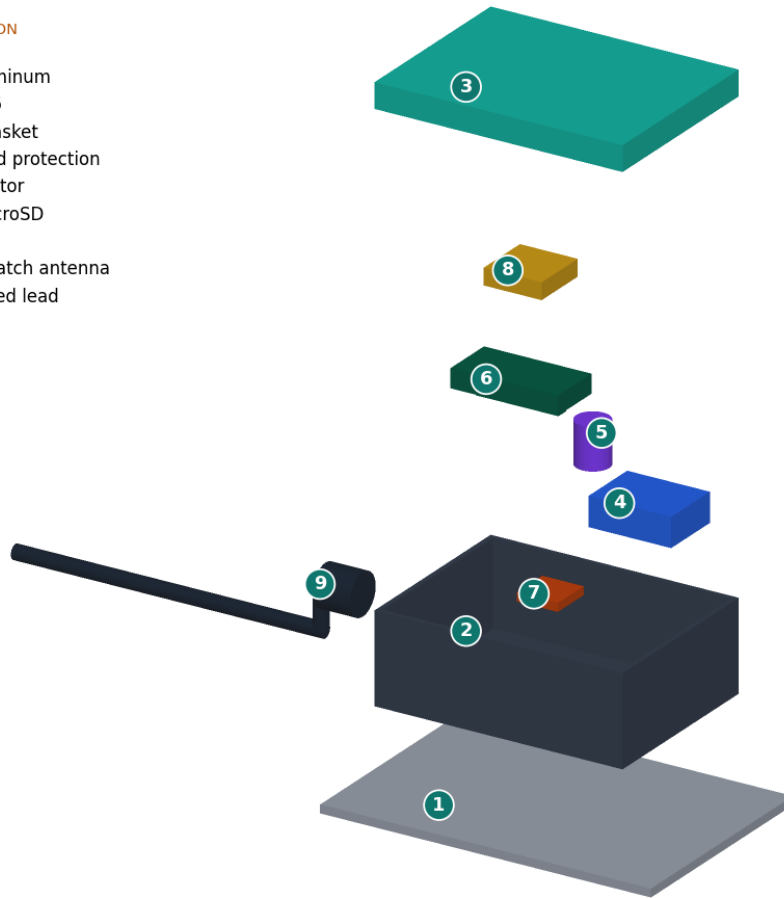


Figure 3. Exploded view with numbered callouts matching the BOM.

PotholeLog: cutaway

CONCEPT, NOT FOR FABRICATION

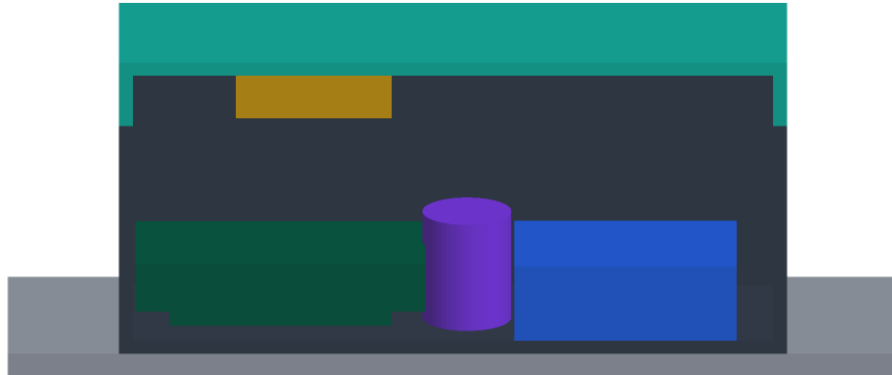


Figure 4. Cutaway. The IMU sits on the enclosure floor, the controller and converter beside it, and the GNSS module under the lid.

■ First-order numbers

All values are estimates for concept review and will be checked at TRL 3.

Table 2. First-order numbers.

Quantity	Estimate	Basis and assumptions	Requirement
Sample spacing on the road	about 35 mm	400 Hz at 50 km/h (13.9 m/s)	R6
Samples across a 300 mm pothole	about 9 at 50 km/h, about 4 at 80 km/h	Length divided by sample spacing	R1, margin thin at high speed
GNSS fix spacing	about 1.4 m	10 Hz at 50 km/h	R4
Raw data rate	about 5.2 kB/s	6 axes x 2 bytes x 400 Hz, plus about 0.4 kB/s of GNSS records	
Raw data per day	about 190 MB	10 h of driving	
Raw storage on a 32 GB card	about 5 months	Rolling buffer	R7 met
Summaries per day	about 0.1 MB	200 km a day, 2,000 segments at 48 bytes, plus up to about 500 events at 32 bytes	R8 met
Running power	about 0.7 W	Controller about 0.35 W with Wi-Fi off, GNSS about 0.1 W, IMU and card about 0.1 W, converter at about 85 % efficiency	R9 met
Supply current	about 60 mA at 12 V, about 30 mA at 24 V	From running power	
Hold-up time	about 9 s	1 F from 5.0 V to 3.5 V gives about 6 J; at 0.7 W	R10 met; closing files needs under 1 s
Size	120 x 90 x 59 mm on a 160 x 110 mm plate	Stock enclosure	
Mass	about 0.4 kg	Plate about 0.19 kg, box about 0.11 kg, electronics and lead about 0.1 kg	
Parts cost	about \$69	Indicative prices, see <code>bom/bom.csv</code>	R15 met with \$1 margin

■ Key design choices

All choices below are proposed, awaiting Amish.

- **Sprung-mass mount on the floor, not on the axle.** The floor is clean, dry and easy to reach, and the GNSS can see the sky through the windows. The axle would give a stronger, less filtered signal but sees large shocks, water and stones. Calibration per vehicle corrects for the suspension.
- **Depot Wi-Fi upload, not cellular.** It keeps parts cost within \$70 and avoids data plans. Cellular (for example an LTE-M module) would give same-day data but adds about \$20 to \$30 per unit and a monthly fee.
- **Raw data kept on the card.** Storage is cheap, and raw data lets the detection method improve without new hardware.
- **Summaries and events, not tracks, leave the vehicle's operator.** Only road-level results are published (R13).
- **Controller.** ESP32-S3 (Wi-Fi, low cost, wide community) is proposed. Alternatives: an RP2040 board with a separate Wi-Fi module, or an nRF52 board with a phone or gateway for upload.

- **Built on existing lab work where it fits.** Outputs are proposed to follow the CityTwin open data export so PotholeLog segments can appear on the CityTwin map; CityTwin is built on TwinKit. PotholeLog does not use FieldNode, because it runs from vehicle power rather than solar and uploads at the depot rather than over LoRaWAN.

■ Safety

Safety: PotholeLog is fitted to vehicles that carry passengers and operate in traffic.

- **Secure mounting.** A loose box can become a projectile in a crash or hard stop. Bolt the plate through with M6 bolts and locking nuts at existing fixing points, never with adhesive or magnets alone. Do not cut, drill or weld chassis members or safety-critical structure; follow the fleet operator's and vehicle maker's rules for body fittings.
- **Vehicle electrics.** Connect only to a fused, ignition-switched circuit through the inline fuse, with the battery isolated during installation. Route the lead away from moving parts, hot surfaces and sharp edges, and protect it where it passes through panels. A short on a 24 V vehicle supply can start a fire.
- **Working on vehicles.** Install only with the vehicle parked, secured and switched off. Never work under a vehicle supported only by a jack.
- **Driver distraction.** The logger has no controls or display and must not be handled while the vehicle is moving.
- **Sharp edges.** Deburr the aluminum plate.
- **Supercapacitor.** It stays charged for a short time after power is removed; do not short its terminals.
- **Data.** Vehicle location data can reveal where drivers are and when. Keep tracks on the operator's server under the operator's data policy, inform drivers and unions before a pilot, and publish only road-level results.

This is a research prototype. It is not certified to automotive electrical or EMC standards, and its outputs must not be used for contract acceptance or safety decisions without validation against a calibrated reference.

■ Open questions for TRL 3

- Which roughness metric from sprung-mass acceleration correlates best with IRI across speed bands, and how many calibration sections does each vehicle need?
- What event threshold and clustering rule meet R1 and R2 together?
- How far off is GNSS in the partner city's densest streets, and is map matching needed for R4?
- Is the 4 mm aluminum plate stiff enough on a typical bus floor to keep the first natural frequency above 150 Hz (R12)?
- Which host fleet and city for a first pilot, and who provides reference IRI?
- Should a bicycle variant be developed, and if so with what battery and mount (R16)?

Concept media: [blueprint sheet](#), [interactive 3D model](#).